EpiBench: Verifiable Evaluation of AI Agents on Epigenetics Analysis

Short-horizon tasks from real epigenetics workflows

Harihara Muralidharan[1], Reema Baskar[1], Soo Hee Lee[1], Kenny Workman[1]

[1]LatchBio, San Francisco, CA, USA

Correspondence: kenny@latch.bio

ABSTRACT

We introduce EpiBench, a verifiable benchmark for short-horizon epigenomics analysis. EpigeneticsBench evaluates if agents can execute well-defined analysis decisions from realistic workflow states and return deterministically gradable answers. The benchmark includes 106 tasks across CUT&Tag/CUT&RUN, ATAC-seq, ChIP-seq, and DNA methylation workflows. Across 5,088 parsed trajectories from 16 model-harness pairs, no system solves a majority of attempts: GPT-5.5 / Pi leads at 151/318 attempts (47.5%), followed by GPT-5.5 / OpenAI Codex at 135/318 attempts (42.5%) and GPT-5.4 / Pi at 134/318 attempts (42.1%). Difficulty is structured by assay type and workflow convention, and component-level scoring reveals frequent partial recovery despite endpoint failure. These results suggest that current agents can often locate and manipulate the right biological evidence, but still fail at brittle local decisions involving metric definitions, comparator scope, normalization state, and answer-surface constraints.

Topline Benchmark Performance

We ran the benchmark across 16 model-harness pairs. Pi denotes the Pi terminal coding harness. Passing requires exact recovery of the graded structured answer for an evaluation attempt. The updated manifest contains 5,088 valid result JSONs in a balanced three-replicate design: 106 evaluations for each of 16 model-harness pairs.

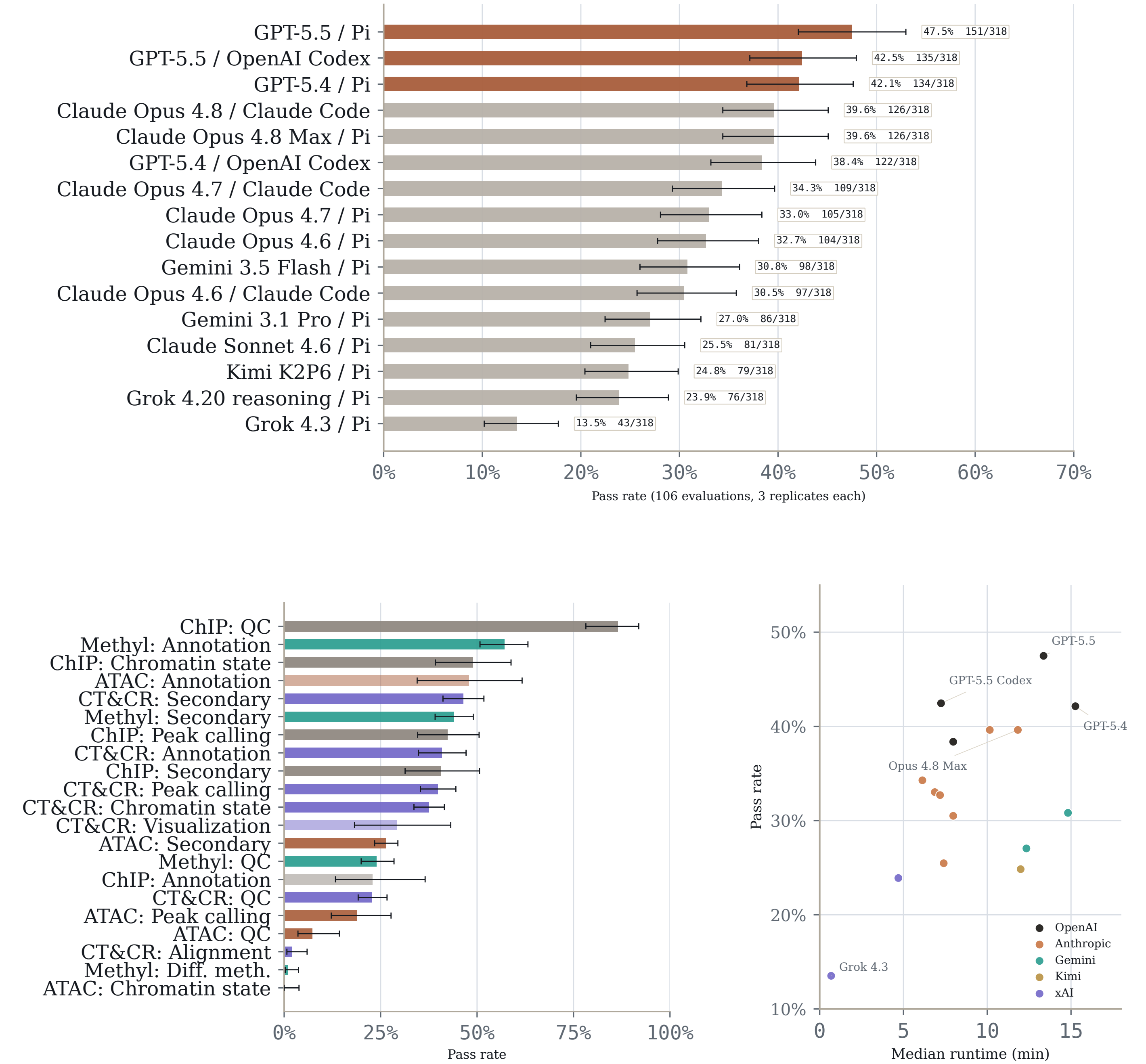

Figure 1: Topline benchmark performance. Top: run-level pass rate by model-harness pair across all parsed attempts. Bottom left: pass rate by assay and task family. Bottom right: pass rate against median runtime. Wilson confidence intervals and evaluation-level replicate statistics are reported in Table 2.

Introduction

Epigenomic assays measure the regulatory state of the genome: chromatin accessibility, histone modifications, DNA methylation, and other molecular signals that link genotype to cell state and gene regulation. But raw epigenomics data are not directly interpretable. Drawing scientific conclusions requires a sequence of analysis decisions across read alignment, peak calling, motif interpretation, sample aggregation, normalization, and genomic annotation. Small procedural choices can change the final biological conclusion.

AI agents show early promise in human-guided analysis of biological data. They can inspect files, write code, invoke command-line tools, and exercise some scientific judgment in analysis tasks. But they remain prone to hallucination and unreliable biological judgment, especially across diverse tissues, diseases, study designs, and assay types.

Existing biology benchmarks increasingly test practical analysis, but, to our knowledge, none focus on epigenomics analysis alone. We introduce EpiBench, a verifiable benchmark for short-horizon epigenomics tasks. Each evaluation snapshots a realistic workflow state immediately before a target result. The agent receives relevant files, metadata, and task context, then must interact with the data and return a structured answer graded deterministically. The benchmark includes 106 evaluations across CUT&Tag/CUT&RUN, ATAC-seq, ChIP-seq, and DNA methylation workflows, covering quality control, peak calling, chromatin-state analysis, genomic annotation, differential methylation, secondary analysis, alignment, and visualization.

Across 5,088 parsed trajectories from 16 model-harness pairs, no system solves a majority of attempts. GPT-5.5 / Pi leads at 151/318 attempts (47.5%), followed by GPT-5.5 / OpenAI Codex at 135/318 attempts (42.5%) and GPT-5.4 / Pi at 134/318 attempts (42.1%). The failures show that current agents can sometimes make reasonable progress with epigenetics data, but still struggle with some analysis decisions, especially those that require biological reasoning: choosing which samples to compare, accurate normalization, identifying appropriate genomic features, and reporting appropriate statistics.

Benchmark Construction

To construct a benchmark that approximates real epigenomics analysis, we derived evaluations from practical workflows across CUT&Tag/CUT&RUN, ATAC-seq, ChIP-seq, and DNA methylation. We decomposed analysis workflows into short, gradeable decisions: quality control, peak calling, chromatin-state comparisons, genomic annotations, methylation statistics, alignment checks, and secondary analyses. At each step, we sought to isolate the empirical decision that determines the result and formalize it as a deterministically gradable answer.

The final evaluation suite consists of 106 problems spanning four assay families and eight task categories. Each problem includes a snapshot of real assay outputs immediately before a target analysis step, metadata needed to interpret the files, a high-level task description, and a deterministic grader that evaluates whether the agent recovered the requested empirical result.

We follow three design criteria. First, tasks must be verifiable: each evaluation admits an automatically checkable success condition, such as a numerical interval check, structured label match, or all-of field comparison. Second, tasks should be scientifically durable: the intended answer should be recoverable across reasonable analysis choices, so the benchmark measures epigenomics reasoning rather than brittle implementation details. Third, tasks should resist shortcuts: agents must interact with the provided data artifacts, and we remove items that can be solved reliably by trivial heuristics or prior biological knowledge alone.

Tasks are designed to specify what should be recovered rather than how it should be recovered. Some procedural evaluations name a specific operation, but most scientific evaluations leave the agent to choose reasonable analysis methods to accomplish a goal. A correct answer should therefore reflect the result supported by the specific assay context, not a nearby generic workflow convention or a plausible biological expectation.

All evaluations underwent manual quality control. We inspected agent trajectories across runs to identify prompts that over-specified the method, answers that could be obtained through shortcuts, and graders that failed to distinguish supported results from plausible but unsupported biological outputs.

Evaluation Inventory

The benchmark covers four assay types and eight task categories. CUT&Tag/CUT&RUN tasks are drawn from zebrafish chromatin workflow snapshots and span chromatin state analysis, QC, peak calling, secondary analysis, genomic annotation, alignment, and visualization. ATAC-seq tasks cover B-ALL and cold-response analyses, with secondary analysis dominating. ChIP-seq tasks include peak calling, chromatin state analysis, QC, genomic annotation, and secondary analysis. Methylation tasks are derived from

GSE149608 and GSE149609, covering QC, secondary analysis, genomic annotation, and differential methylation.

Assay type	Evals	Sources	Task categories
CUT&Tag/CUT&RUN	47	Zebrafish chromatin workflow snapshots	Chromatin state (12), QC (10), peak calling (9), secondary analysis (7), genomic annotation (5), alignment (3), visualization (1)
ATAC-seq	24	B-ALL ATAC, cold-response ATAC	Secondary analysis (17), chromatin state (2), peak calling (2), QC (2), genomic annotation (1)
ChIP-seq	10	ChIP-seq workflow snapshots	Peak calling (3), chromatin state (2), QC (2), secondary analysis (2), genomic annotation (1)
Methylation-seq	25	GSE149608, GSE149609	QC (8), secondary analysis (8), genomic annotation (5), differential methylation (4)

Table 1: Evaluation inventory. The benchmark groups evaluations by assay type while varying the source workflow, task category, and expected answer surface within each group.

Across all 106 evaluations, 34 are secondary analysis tasks, 22 are QC, 16 are chromatin state analysis, 14 are peak calling, 12 are genomic annotation, 4 are differential methylation, 3 are alignment, and 1 is visualization. The updated result manifest contains 5,088 valid trajectories from 16 model-harness pairs, with exactly 318 attempts per model-harness pair.

Results

Top agents remain below 50% endpoint accuracy

Across 16 model-harness pairs and 5,088 valid trajectories, deterministic final-answer grading produced 1,672 passes (32.9%). No system reached 50% endpoint accuracy. GPT-5.5 / Pi was the descriptive leader, passing 151/318 attempts (47.5%; Wilson 95% confidence interval (CI), 42.1–53.0), followed by GPT-5.5 / OpenAI Codex at 135/318 (42.5%; 37.1–47.9) and GPT-5.4 / Pi at 134/318 (42.1%; 36.8–47.6). Claude Opus 4.8 Max / Pi and Claude Opus 4.8 / Claude Code both passed 126/318 attempts (39.6%).

The updated manifest is balanced: each model-harness pair contributes three rounds for each of 106 evaluations, so differences reflect the same evaluation set rather than coverage differences. The top systems form a leading band rather than a clean ordering: several Wilson intervals overlap, and the gap between second and third place is only one passing run. We therefore treat this section as a measurement of current endpoint accuracy and reliability, not as a finely resolved leaderboard.

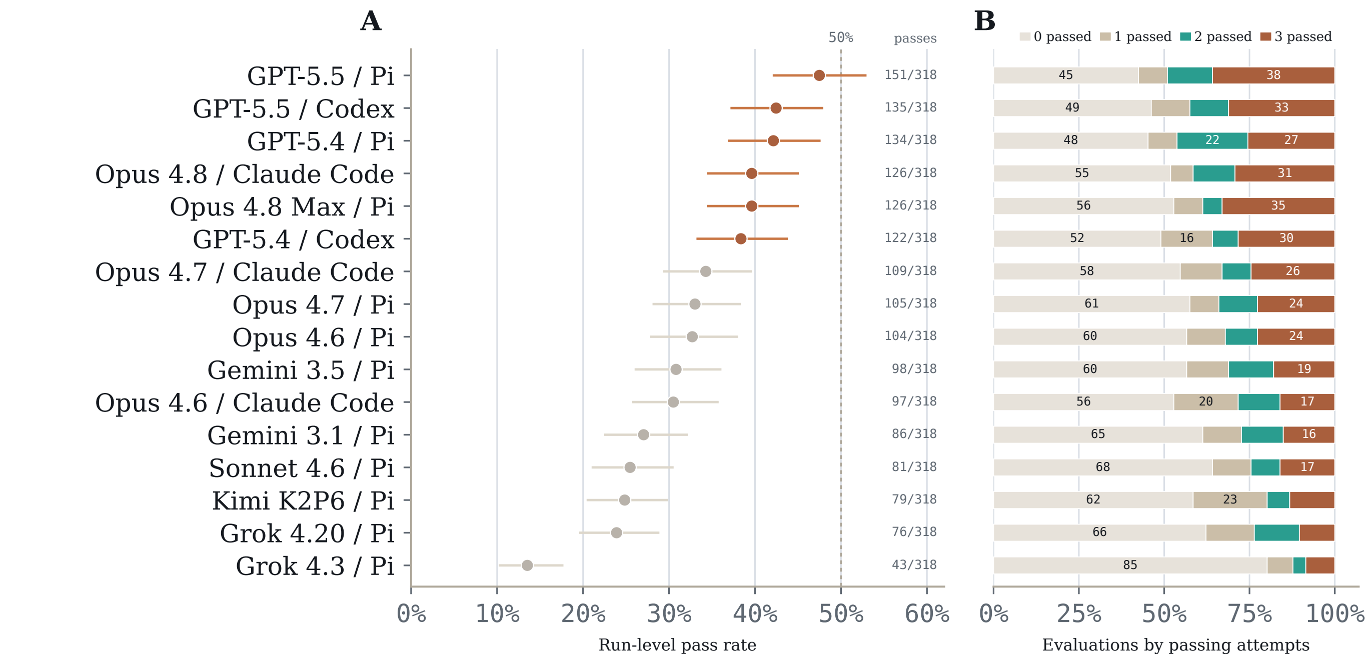

Figure 2: Endpoint accuracy and replicate outcomes across model-harness pairs. Panel A: run-level pass rate across all 5,088 parsed attempts, with Wilson confidence intervals and a 50% reference line. Panel B: evaluation-level replicate outcomes over the same model order, partitioning the 106 evaluations for each model-harness pair by whether zero, one, two, or all three attempts passed.

Model / harness	Pass runs	Pass %	Wilson 95% CI	Any	Majority	All
GPT-5.5 / Pi	151/318	47.5	42.1–53.0	61/106	52/106	38/106
GPT-5.5 / OpenAI Codex	135/318	42.5	37.1–47.9	57/106	45/106	33/106
GPT-5.4 / Pi	134/318	42.1	36.8–47.6	58/106	49/106	27/106
Claude Opus 4.8 Max / Pi	126/318	39.6	34.4–45.1	50/106	41/106	35/106
Claude Opus 4.8 / Claude Code	126/318	39.6	34.4–45.1	51/106	44/106	31/106
GPT-5.4 / OpenAI Codex	122/318	38.4	33.2–43.8	54/106	38/106	30/106
Claude Opus 4.7 / Claude Code	109/318	34.3	29.3–39.7	48/106	35/106	26/106
Claude Opus 4.7 / Pi	105/318	33.0	28.1–38.4	45/106	36/106	24/106
Claude Opus 4.6 / Pi	104/318	32.7	27.8–38.0	46/106	34/106	24/106
Gemini 3.5 Flash / Pi	98/318	30.8	26.0–36.1	46/106	33/106	19/106
Claude Opus 4.6 / Claude Code	97/318	30.5	25.7–35.8	50/106	30/106	17/106
Gemini 3.1 Pro / Pi	86/318	27.0	22.5–32.2	41/106	29/106	16/106
Claude Sonnet 4.6 / Pi	81/318	25.5	21.0–30.5	38/106	26/106	17/106
Kimi K2P6 / Pi	79/318	24.8	20.4–29.9	44/106	21/106	14/106
Grok 4.20 reasoning / Pi	76/318	23.9	19.5–28.9	40/106	25/106	11/106
Grok 4.3 / Pi	43/318	13.5	10.2–17.7	21/106	13/106	9/106

Table 2: Deterministic final-answer performance. Run-level pass rates are computed over the balanced updated manifest. Any, Majority, and All report evaluation-level replicate statistics over 106 evaluations for each model-harness pair. Runtime and cost are treated as operational diagnostics rather than leaderboard columns.

Assay type separates endpoint performance

Endpoint accuracy changed sharply across assays. ChIP-seq tasks had the highest run-level pass rate, with 241/480 passing attempts (50.2%), followed by CUT&Tag/CUT&RUN at 768/2,256 (34.0%), methylation-seq at 400/1,200 (33.3%), and ATAC-seq at 263/1,152 (22.8%). The ChIP-seq estimate should be read cautiously because it comes from only 10 evaluations, but the aggregate pattern is still useful: agents are not failing epigenetics analysis uniformly. They are more reliable in some assay representations than in others.

The assay differences reflect workflow conventions as much as biological domain. ATAC-seq was the hardest aggregate despite familiar peak-level data, because many tasks required preserving the correct sample, peak, or contrast definition through secondary analysis. CUT&Tag/CUT&RUN occupied the middle of the distribution, with stronger performance on peak calling, annotation, and secondary analysis than on alignment or QC. Methylation-seq also sat near the middle overall, but it mixed relatively tractable annotation and secondary analysis tasks with brittle coverage, strand, and effect-size decisions. ChIP-seq was easiest in this inventory, driven in part by high performance on quality-control tasks.

The assay inventory is broad but not balanced. CUT&Tag/CUT&RUN contributes 47 evaluations, methylation-seq 25, ATAC-seq 24, and ChIP-seq 10 (Figure 3). We therefore use assay-level performance as the main result and use the assay-by-task matrix only as a diagnostic check on which assay-specific cells drive each aggregate estimate.

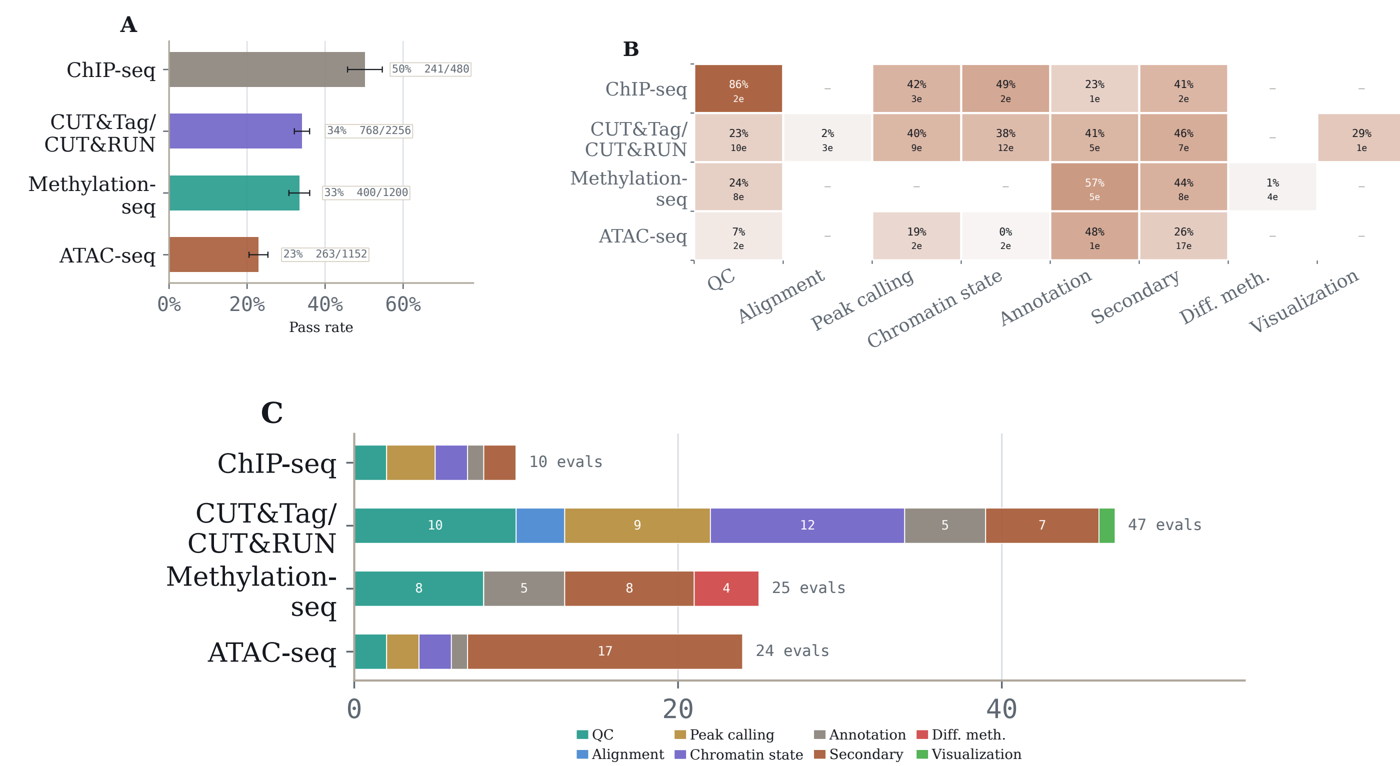

Figure 3: Performance varies by assay type. Panel A: run-level pass rate by assay type with Wilson confidence intervals and passing-attempt counts. Panel B: assay-by-task-category pass-rate heatmap; cell text reports pass rate and the number of evaluations in that cell. Panel C: evaluation-count composition by assay type, showing the task mix behind each assay summary.

Endpoint failures often preserve partial work

Replicate statistics separate reachable success from stable success. Averaged over model-harness pairs, agents passed at least one replicate on 44.2% of evaluations, a majority of replicates on 32.5%, and all three replicates on 21.9%. The leading GPT-5.5 / Pi run shows the same pattern: 61/106 evaluations had at least one passing attempt, 52/106 had majority recovery, and only 38/106 passed all three attempts. A single passing trajectory therefore indicates reachable capability, but not repeatable operation.

Component scores explain much of this instability. Among 5,059 trajectories with component-field scores, endpoint pass/fail exactly matched whether every required field passed. Individual fields passed at 69.6%, more than twice the 32.9% endpoint pass rate. The hardest evaluations were not uniformly blank failures: 21 evaluations had no passing attempts, but 17 of those had nonzero mean grader scores and 9 had field-level pass rates of at least 50%. Agents often recovered the right evidence or an adjacent statistic, then missed one required field, comparator, denominator, or answer-surface constraint.

This pattern is not primarily explained by execution crashes. No parsed trajectory reported a timeout or detected out-of-memory event. Recorded runtimes varied widely (median 506 seconds; 90th percentile 2,351 seconds), but the dominant error mode in this manifest is a brittle final analysis decision rather than an inability to run code.

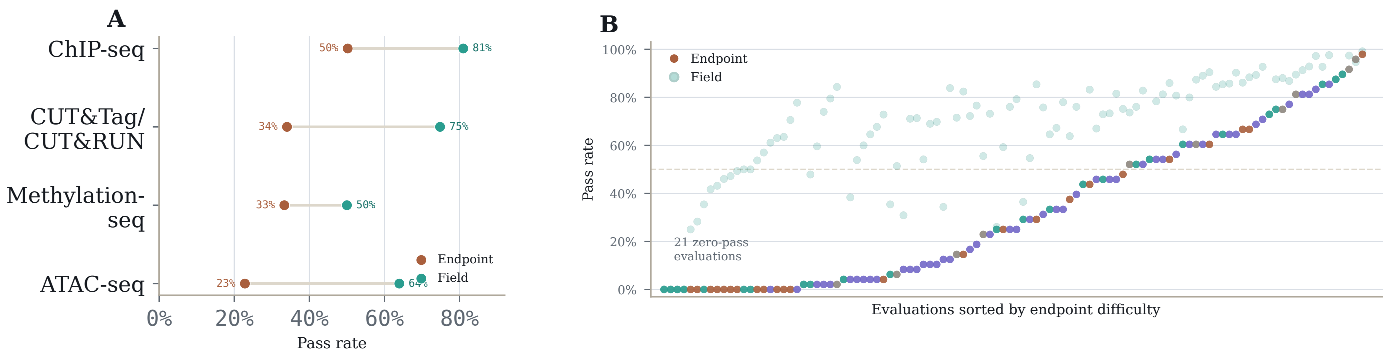

Figure 4: Endpoint failures contain substantial partial progress. Panel A: endpoint and component-field pass rates by assay type. Panel B: evaluation-level difficulty spectrum, with endpoint and field-level pass rates for each evaluation. Field scores are diagnostics, not replacement benchmark scores.

Errors trace to assay-specific choices

Manual review localizes endpoint misses to concrete analysis decisions. Review coverage is incomplete (25/106 evaluations), and the labels are task-level diagnostics rather than per-run error calls. Within that reviewed subset, the most common error types were incorrect statistic or ranking (14 evaluations; 33.9% pass rate), unstable threshold choice (9; 31.9%), biological prior over provided evidence (8; 45.6%), gene/peak identifier mismatch (6; 0.0%), and incorrect biological unit or reference set (6; 42.7%). These are not generic reasoning failures. They are local choices about which data layer, feature identity, comparison group, threshold, or biological prior should control the final answer.

The reviewed traps also explain why evaluation difficulty is correlated. One missing assay convention can sink several tasks at once. In CUT&RUN spike-in normalization, choosing Bowtie2 end-to-end alignment instead of the appropriate local mode can discard most contaminated spike-in reads and corrupt every downstream normalization result. In WGBS outputs, failing to merge the two Bismark rows representing one CpG dinucleotide changes coverage and downstream differential-methylation calls. In ATAC-seq, feeding paired-end BAMs directly to MACS3 with `-f BAMPE` can replace the intended Tn5 insertion-site workflow and propagate into peak, motif, and integration analyses.

Prior-driven assertion is a separate recurring pattern. Agents sometimes substitute well-known regulatory biology for the calculation supported by the provided files: promoter methylation instead of the requested gene-body comparison, canonical driver genes instead of the computed enhancer ranking, or differentiated-cell references instead of the stem-cell reference required for a Polycomb methylation enrichment. These examples frame the benchmark's core measurement target: not whether an agent can invoke bioinformatics tools, but whether it can commit to the assay-specific decision that makes the computed result verifiable.

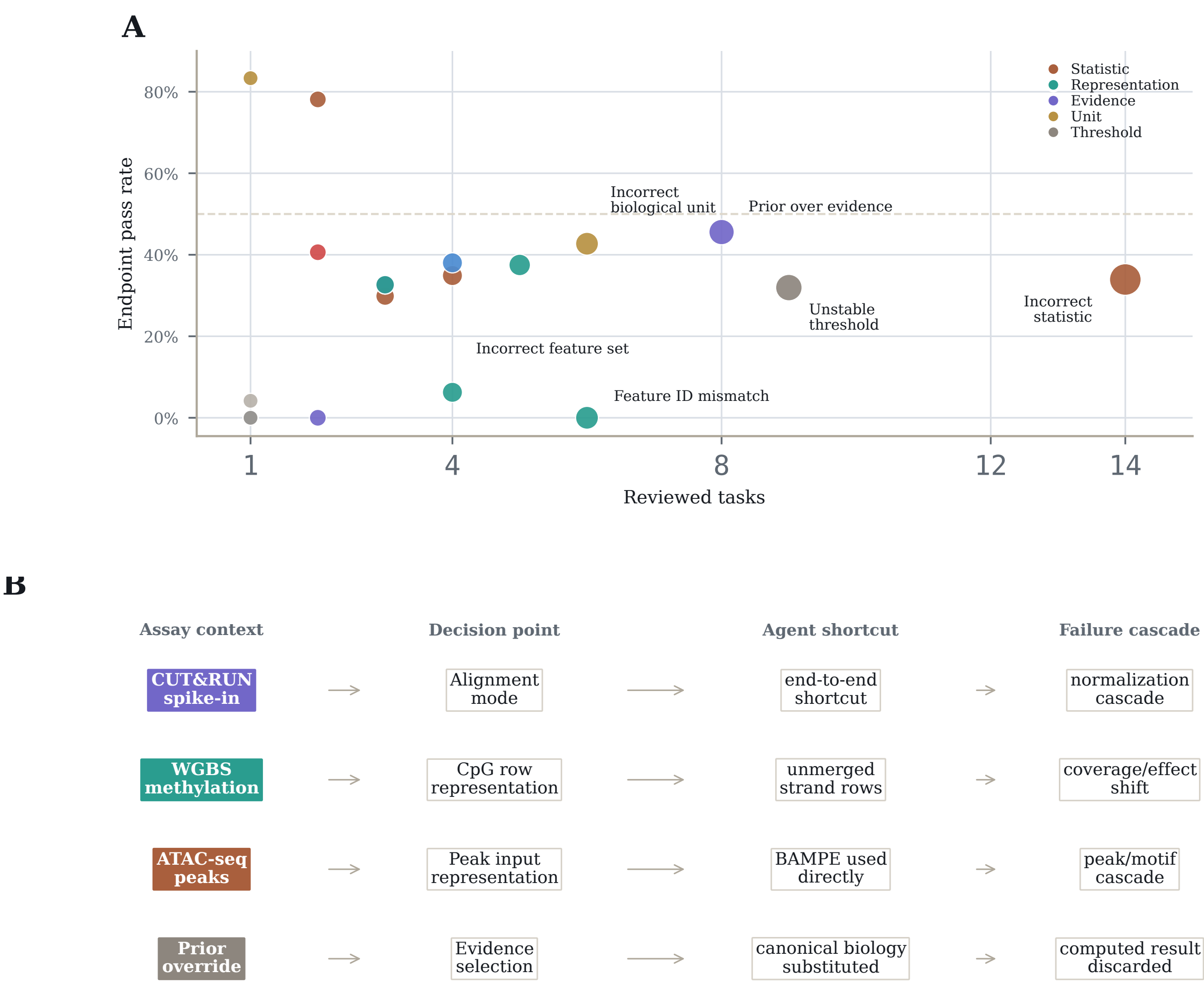

Figure 5: Reviewed failures concentrate in assay-specific decision traps. Panel A: frequency and severity of recurring error types from manual review; each point is an error type, with lower pass rates indicating harder traps. Panel B: representative assay decisions showing how a local shortcut can propagate into downstream endpoint failure. Manual review covers 25/106 evaluations and should be interpreted as diagnostic rather than exhaustive.

Discussion

The short-horizon epigenomics benchmark occupies a middle ground between localized workflow execution and long-horizon scientific reasoning. It is narrower than SpatialBench-Long because each task targets a focused result, but it is more empirical than biological question answering because the answer depends on interacting with real assay outputs.

The main result is not that agents fail universally. Leading systems recover roughly half of available endpoint answers, and many failures show substantial component-level progress. The limitation is reliability: even strong model-harness pairs remain sensitive to representation, metric definition, and comparator choice. These are precisely the local decisions that determine whether an epigenomics analysis is trustworthy.

The trap taxonomy clarifies what the benchmark measures. Agents can run Bowtie2, MACS2, Picard, and HOMER correctly — they fail at choosing `-local` vs `end-to-end`, `-f BAMPE` vs `-f BAM`, `-broad` vs narrow, or the appropriate `-keep-dup` setting. The bottleneck is domain-specific parameter selection that requires understanding *why* the tool behaves differently for a given mark or assay, not *how* to invoke it. This positions the benchmark as a test of scientific judgment rather than tool proficiency.

Three cross-assay failure patterns generalize beyond epigenomics: (1) prior-driven assertion — asserting from training data rather than computing from provided files; (2) depth and normalization confounding — equalizing after feature detection rather than before; and (3) format/mode selection — the same tool flag producing opposite errors depending on assay context (e.g., `-f BAMPE` inflates CUT&Tag peaks by 25–75% but deflates ATAC-seq peaks by $\sim 10\times$). These are not epigenomics-specific; any scientific benchmark involving bioinformatics tools will encounter agents that know tool syntax but not parameter semantics.

The progressive-difficulty design is validated by the trap analysis. Procedural evaluations saturate at the ceiling model (100% pass, requiring hardening), scientific-judgment evaluations discriminate (30–50% pass at ceiling), and conceptual-biology evaluations defeat all models (0% pass, probing future capability). As weaker models are tested, failures should migrate from the decision layer ("computed correctly but chose wrong") to the computation layer ("cannot compute at all"), producing the graduated difficulty curve needed for meaningful model separation.

The benchmark has several limitations. The updated manifest is balanced across model-harness pairs, but the task inventory is not balanced across assay families: CUT&Tag/CUT&RUN (47 evaluations) and methylation-seq (25) contribute more evaluations than ChIP-seq (10), and secondary analysis plus QC tasks dominate the inventory. The correlation structure of trap clusters means aggregate scores partially reflect repeated tests of the same underlying decision; this mirrors real workflow failure modes but should be acknowledged when interpreting per-evaluation independence. Finally, deterministic graders intentionally constrain the answer surface; they are useful for measurement, but they do not capture every scientifically valid analysis path.

Methods

Evaluation construction

Candidate evaluations were derived from real epigenomics analysis workflows. An evaluation was retained when the target result could be reproduced from the provided data, expressed through a constrained final answer, and graded deterministically. Tasks were revised or removed when prompts over-specified the method, when reasonable analysis choices produced incompatible answers, or when the grader could not distinguish a supported result from plausible but unsupported biological output.

Agent runs and aggregation

Each model-harness pair was run with up to three replicates per evaluation. The updated manifest used here is `epibench_manifest_updated.csv`; the full SHA-256 hash is recorded in the derived summary facts table. All 5,088 listed result JSONs were downloaded and parsed successfully. The manifest contains a complete three-replicate design for 106 evaluations and 16 model-harness pairs. Failed, malformed, or nonpassing endpoint grader outputs are counted as nonpassing attempts when present; no downloaded result JSON was missing or invalid in this analysis.

Statistical analysis

The primary metric is run-level pass rate. Wilson binomial confidence intervals summarize run-level uncertainty and do not account for evaluation-level clustering. We therefore also report evaluation-level replicate summaries: whether a model-harness pair passed any attempt, a majority of attempts, or all three attempts for an evaluation. Because the updated manifest is balanced, run-level and balanced-design summaries are identical. Component field scores and failure-mode labels are reported as diagnostics rather than benchmark scores.

Data availability

Results files, public example evaluations, and representative trajectories will be made available at `https://github.com/latchbio/epigenetics-short-horizon-benchmark`. The derived summary tables used for this manuscript are in `analysis_outputs/epi_manifest_updated_summary/`.

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